







# STATE TECHNICAL UNIVERSITY

END SEMESTER EXAMINATION 2025-2026 | ANSWER BOOKLET

CANDIDATE ROLL NO: 2026-CS-PH-050  
CANDIDATE NAME: Rajat Mittal  
EXAMINATION / COURSE: B.Tech Semester II  
SUBJECT & CODE: Physics (CS 201)  
BOOKLET SERIAL NO: BK-PH-80050

BARCODE SCAN TRACKING



\*BK-PH-80050\*

| Q.No  | Q1 | Q2 | Q3 | Q4 | Q5 |
|-------|----|----|----|----|----|
| Max   | 14 | 14 | 14 | 14 | 14 |
| Marks |    |    |    |    |    |

START WRITING YOUR ANSWERS BELOW. DO NOT WRITE IN THE MARGINS.

## Q.1 Ans 1. Group Velocity and Particle Velocity in Quantum Mechanics:

According to de Broglie, matter exhibits dual wave-particle properties. A localized particle is represented by superposition of plane waves forming a wave packet:

$$\Psi(x, t) = \text{Integral} [A(k) \cdot \exp(i(kx - \omega t)) dk]$$

- Phase velocity:  $v_p = \omega / k$ . For free particle,  $E = \hbar \omega = p^2 / 2m \Rightarrow \omega = \hbar k^2 / 2m$ .

$$v_p = \hbar k / 2m = p / 2m = v / 2.$$

- Group velocity:  $v_g = d(\omega) / dk = d/dk [\hbar k^2 / (2m)] = \hbar k / m = p / m = v$ .

Significance: The probability density  $|\Psi(x, t)|^2$  envelope propagates at  $v_g$ . Individual phase waves travel at  $v_p = c^2 / v > c$  in relativistic frames, but energy and information travel strictly at  $v_g = v \leq c$ .







**Q.2***Ans 2. Wave Optics - Interference of Light:**Interference is the redistribution of luminous energy due to superposition of two coherent wave trains:*

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos(\delta).$$

*Conditions for Sustained Interference:*

1. Sources must be strictly coherent (constant phase difference).
2. Sources must be monochromatic (single wavelength  $\lambda$ ).
3. Equal or nearly equal amplitudes for maximum fringe contrast.

*Division of Wavefront vs Division of Amplitude:**a) Division of Wavefront:*

- Original wavefront is divided into two parts by mirrors, prisms or slits.
- Examples: Young's Double Slit, Fresnel Biprism, Lloyd's Mirror.
- Requires a point or narrow slit source. Fringe width  $\beta = \lambda D / d$ .

*b) Division of Amplitude:*

- Incident wave amplitude is split by partial reflection and refraction at interfaces.
- Examples: Thin dielectric films, Newton's Rings, Michelson Interferometer.
- Can use broad extended sources, producing high intensity fringes.
- Path difference in thin film:  $\Delta = 2 \mu t \cos(r) - \lambda / 2$  (Stokes' phase shift).
- Applications: Anti-reflection optical coatings ( $t = \lambda / 4\mu$ ) and optical surface testing.







**Q.3***Ans 3. Lasers and Fibre Optics:**1. Laser Principles (Light Amplification by Stimulated Emission of Radiation):*

- *Stimulated Absorption: Ground atom absorbs  $h\nu$  and jumps to  $E_2$ .*
- *Spontaneous Emission: Random downward transition emitting photon: Rate =  $A_{21} * N_2$ .*
- *Stimulated Emission: Incident photon induces excited atom to emit an identical twin photon (same frequency, phase, direction): Rate =  $B_{21} * N_2 * \rho(\nu)$ .*
- *Population Inversion ( $N_2 > N_1$ ): Non-equilibrium state achieved via optical/electrical pumping in a 3-level or 4-level system with a metastable state (lifetime  $\sim 10^{-3}$  s).*
- *Optical Resonator: Pair of mirrors (100% and 95% reflection) creating standing waves:  $L = m * \lambda / 2$ .*

*2. Fibre Optics Fundamentals:*

- *Wave propagation occurs by Total Internal Reflection (TIR) at Core-Cladding interface ( $n_1 > n_2$ ).*
- *Critical Angle:  $\theta_c = \sin^{-1}(n_2 / n_1)$ .*
- *Acceptance Angle ( $\theta_0$ ): Maximum entry angle:*  

$$\sin(\theta_0) = \sqrt{n_1^2 - n_2^2} = NA \text{ (Numerical Aperture).}$$
- *Fractional Refractive Index Change:  $\Delta = (n_1 - n_2) / n_1$ .*
- *$NA = n_1 * \sqrt{2 * \Delta}$ .*
- *Types: Step Index (Single mode / Multimode) and Graded Index (GRIN) fibers with parabolic index profile reducing intermodal dispersion.*







**Q.4***Ans 4. Electromagnetism - Maxwell's Equations & Boundary Conditions:**1. Maxwell's Four Fundamental Equations:**a) Gauss's Law for Electrostatics:*

$$\text{Differential: } \operatorname{div}(\mathbf{D}) = \rho_v \quad / \quad \text{Integral: } \oiint \mathbf{D} \cdot d\mathbf{A} = Q_{\text{enclosed}}$$

*Physical meaning: Electric flux through closed surface equals total enclosed charge.**b) Gauss's Law for Magnetism:*

$$\text{Differential: } \operatorname{div}(\mathbf{B}) = 0 \quad / \quad \text{Integral: } \oiint \mathbf{B} \cdot d\mathbf{A} = 0$$

*Physical meaning: Magnetic monopoles do not exist; magnetic lines form continuous loops.**c) Faraday's Law of Electromagnetic Induction:*

$$\text{Differential: } \operatorname{curl}(\mathbf{E}) = -d\mathbf{B}/dt \quad / \quad \text{Integral: } \oint \mathbf{E} \cdot d\mathbf{l} = -d(\Phi_B)/dt$$

*Physical meaning: Time-varying magnetic fields induce circulating electric fields.**d) Ampere-Maxwell Law:*

$$\text{Differential: } \operatorname{curl}(\mathbf{H}) = \mathbf{J} + d\mathbf{D}/dt \quad / \quad \text{Integral: } \oint \mathbf{H} \cdot d\mathbf{l} = I_{\text{cond}} + d(\Phi_E)/dt$$

*Displacement current density:  $\mathbf{J}_d = d\mathbf{D}/dt = \epsilon \cdot d\mathbf{E}/dt$ , resolving continuity equation  $\operatorname{div}(\mathbf{J}) + d\rho/dt = 0$ .**2. Boundary Conditions at Dielectric Interfaces:*

- Tangential E-field is continuous:  $E_{1_t} = E_{2_t}$ .
- Normal D-field discontinuity:  $D_{1_n} - D_{2_n} = \rho_s$  (if  $\rho_s = 0$ ,  $D_{1_n} = D_{2_n}$ ).
- Tangential H-field continuity (for non-conducting boundary):  $H_{1_t} = H_{2_t}$ .
- Normal B-field is always continuous:  $B_{1_n} = B_{2_n}$ .

*3. Poynting Vector:  $\mathbf{S} = \mathbf{E} \times \mathbf{H}$  ( $\text{W/m}^2$ ), representing instantaneous directional power flow per unit area.*







**Q.5***Ans 5. Solid State & Semiconductor Physics:**1. Crystal Structures and Lattice Geometry:*

- Simple Cubic (SC): Atoms at 8 corners, Coordination No (CN) = 6, Atomic Packing Factor (APF) = 52%.
- Body Centered Cubic (BCC): Atoms at corners + body center, CN = 8, APF = 68%.
- Face Centered Cubic (FCC): Atoms at corners + face centers, CN = 12, APF = 74%.
- Miller Indices (h k l): Reciprocal of planar intercepts cleared of fractions. Interplanar distance:  $d = a / \sqrt{h^2 + k^2 + l^2}$ .

*2. Energy Band Theory:*

- Kronig-Penney Model explains periodic lattice potential causing splitting of energy levels into allowed bands and forbidden gaps ( $E_g$ ).
- Conductors ( $E_g = 0$  eV), Semiconductors ( $E_g \sim 1.1$  eV for Si), Insulators ( $E_g > 5$  eV).

*3. Intrinsic vs Extrinsic Semiconductors:*

- Intrinsic (pure Si):  $n_i = p_i = \sqrt{N_c \cdot N_v} \cdot \exp(-E_g / 2kT)$ . Fermi level  $E_f$  lies in middle of bandgap.
- n-type: Doped with pentavalent donors (P, As); majority carriers electrons;  $E_f$  shifts close to conduction band.
- p-type: Doped with trivalent acceptors (B, Al); majority carriers holes;  $E_f$  shifts close to valence band.

*4. Direct vs Indirect Bandgap:*

- Direct (e.g. GaAs): Conduction band minimum & valence band maximum align at  $k = 0$  ( $\Delta k = 0$ ); efficient radiative recombination (ideal for LEDs/lasers).
- Indirect (e.g. Si): Peak and valley are misaligned; requires phonon participation, releasing heat.

--- End of Answer Booklet (All 5 questions attempted) ---

Candidate Sign: Rajat